In-class Template

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1. Set up

Packages

```r
# general use and cleaning
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(here)
```

```
here() starts at /Users/nataliechen/git/week-05_spring-2026_aquatic-inverts
```

```r
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
library(snakecase)

# wrap axis labels
library(scales)
```

```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```r
# read in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```r
# reads from taxon list from the data file
taxon_list <- read_csv(here("data","taxon_list.csv"))
```

```
Rows: 50 Columns: 15
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (15): verbatim_name, taxonRank, scientificName, kingdom, Phylum, Class, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(here("data","Aquatic Sampling Data-2026-03-10.xlsx"), sheet = "Aquatic In

# site info
sites <- read_xlsx(here("data","Aquatic Sampling Data-2026-03-10.xlsx"), sheet = "sites")

# water quality

water_quality <-read_xlsx(
  here("data","Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("","n/a","N/A","over","173+"))
```

2. Cleaning

`sites` object

Goal: Create a new object from 'sites" that only includes NCOS sites

```
# creating new object from sites
ncos_sites <- sites |>

  # filter to only include NCOS quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

clean `taxon_list`

Goal: convert character strings in the existing `taxon_list` data frame into snakecase

```
# creating new object from taxon_list
taxon_list_clean <- taxon_list |>

  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

clean `water_quality`

Goal: create a data frame with median pH, median DO (mg/L), median salinity (ppt) for each date and site

pseudocode

- create new object from `water_quality`
- clean column names using janitor
- only include surveys from quarterly NCOS sites
- create new column called `date_site` to join w later data frames
- group by date_site
- calculate median pH, salinity, and dissolved oxygen
- ungroup
- filter out any outliers

```
water_quality_clean <- water_quality |>

  clean_names() |>

  semi_join(sites, by = "site") |>

  # combine date and site together
  unite("date_site", date, site, remove = TRUE) |>

  # group by date_site
  group_by(date_site) |>

  # create new columns that show median of each variable at each date_site
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>

  ungroup() |>

  # filter out salinity outliers
  filter(date_site != "2022-08-12-NVBR")
```

clean `aq_ins`

Goal: create a long format data frame in which each row contains the counted individuals in each taxon during each survey. The data frame should include taxonomic info for each organism and the water quality info for each survey

- create a new object from `aq_ins`
- clean column names
- filter to only include quarterly NCOS sites
- select columns
- filter to only include "filtered beaker" sample
- convert the data frame to long format
- gorup by site, date, and taxon
- calculate average abundance per liter (sum counts divided by 7.5 liters)
- create a new column called `data_site` to join w water quality
- join data frames
- join w taxon list

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>

  clean_names() |>

  semi_join(ncos_sites, by = "site") |>

  rename(date = date_on_vial) |>

  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>

  filter(sample_type %in% c("FB250", "FB 250"))  |>

  pivot_longer(
    # cols = columns to make longer
    cols = ostracod:annelida_polychaete,
    names_to = "taxon",
    values_to = "abundance"
  ) |>

  group_by(date, site, taxon) |>
```

```r
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>

  ungroup() |>

   # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>

  # join with water quality
  left_join(water_quality_clean, by = "date_site") |>

  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>

  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by date, site, and taxon.
i Output is grouped by date and site.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(date, site, taxon))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```